

PAPER_678: UQFF LISA vs LIGO Cross-Sensitivity Comparison and Crossover Frequency

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Framework: Unified Quantum Field Framework (UQFF) v5.30

Session: 173

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Class: LISAVsLIGOComparisons | CP4 #262

Source: grok_share_fc21e30c24b4.txt

Domain: Gravitational Wave Detectors

Abstract

We compute the UQFF suppression ratio $R_{\text{supp}}(f) = h_{\text{UQFF}}(f)/h_{\text{GR}}(f)$ across both detector bands. A crossover frequency f_{cross} exists where LISA-band and LIGO-band suppressions are equal: dominated by $S_{\text{UA,LISA}}$ at low f and by S_{U_m} at high f . We show that UQFF corrections are $\sim 10\%$ larger in the LISA band than in the LIGO band for stellar-mass BH sources.

Primary UQFF Equation

$$R_{\text{supp}}(f) = (1 - f_{\text{TRZ}}) \cdot S_{\text{SCm}} \cdot \min(S_{U_m}, S_{\text{UA,LISA}})$$

Parameters

Band	Freq Range	Dominant UQFF Effect
LIGO	10-2000 Hz	S_{U_m}
LISA	10^{-4} -1 Hz	$S_{\text{UA,LISA}}$

UQFF Constants

Constant	Value	Description
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	Universal Aether density
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "LISAVsLIGOComparisons.h"
```

```
LISAVsLIGOComparisons obj;  
auto result = obj.compute_primary(...); // primary equation  
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import LISAVsLIGOComparisonsCalculator
```

```
calc = LISAVsLIGOComparisonsCalculator()  
result = calc.compute() # returns all UQFF quantities  
print(result)
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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

LIGO O3 sensitivity 2021; Amaro-Seoane 2017 LISA; UQFF Session 173

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